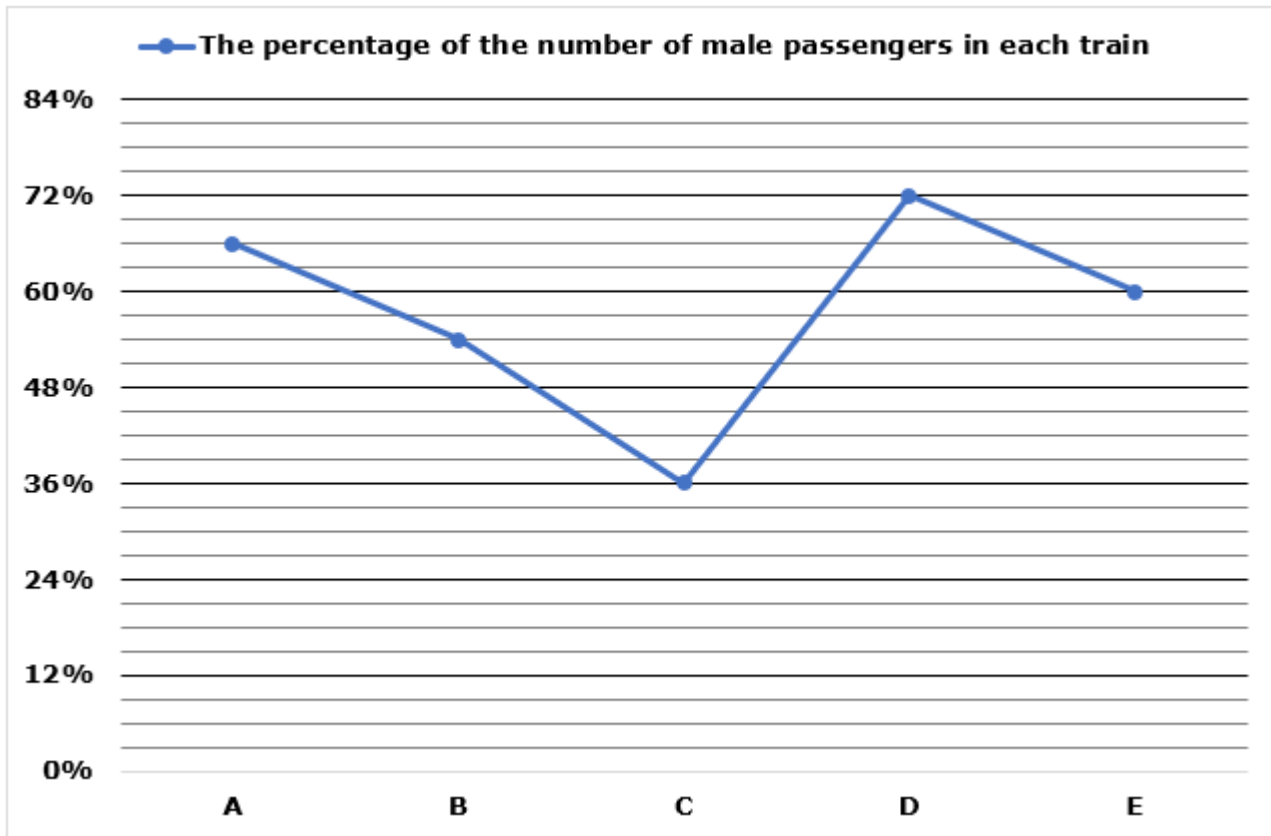


1. Questions

Study the following information carefully and answer the following questions.

The line graph below shows the percentage of the number of male passengers in each train out of the total number of passengers in each train.



The table below shows the number of coaches and the number of passengers in each coach in five different trains.

Trains	The number of coaches	The number of passengers in each coach
A	3X	50
B	20	8X
C	25	-
D	12	15X
E	10	60

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Notes:

- The total number of passengers in a train = The number of coaches * The number of passengers in each coach
- The total number of passengers in a train = the number of male passengers + the number of female passengers
- The number of female passengers in train B is 113 more than that of train A.

If two new coaches, each with a seat capacity of $15X$, are added to Train A and the number of male passengers increases by 20%, find the ratio of the number of male passengers to the number of female passengers in Train A now.

- a. 32: 15
- b. 33: 17
- c. 11: 7
- d. 15: 7
- e. 32: 17

2. Questions

If the number of male passengers is 217 less than the number of female passengers in train C. The number of passengers in each coach in train C is what percentage of that in train B?

- a. 77.5%
- b. 72.5%
- c. 75%
- d. 82.5%
- e. 67.5%

3. Questions

If the ratio of the number of AC coaches to the sleeper coaches in train B is 1:4, respectively. If the number of male passengers in AC coaches in train B is $5X\%$ out of the total male passengers in train B. Find the number of female passengers in sleeper coaches in train B.

- a. 324
- b. 108
- c. 316
- d. 256
- e. Cannot be determined

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4. Questions

If the total number of passengers in train C is $5X\%$ more than that of train E, find the number of passengers in each coach in train C is how much more or less than that in train E.

I: $6X$

II: 30

III: $X + 25$

- a. Both I and II

- b. Only II
- c. Only I
- d. All I, II, and III
- e. Both II and III

5. Questions

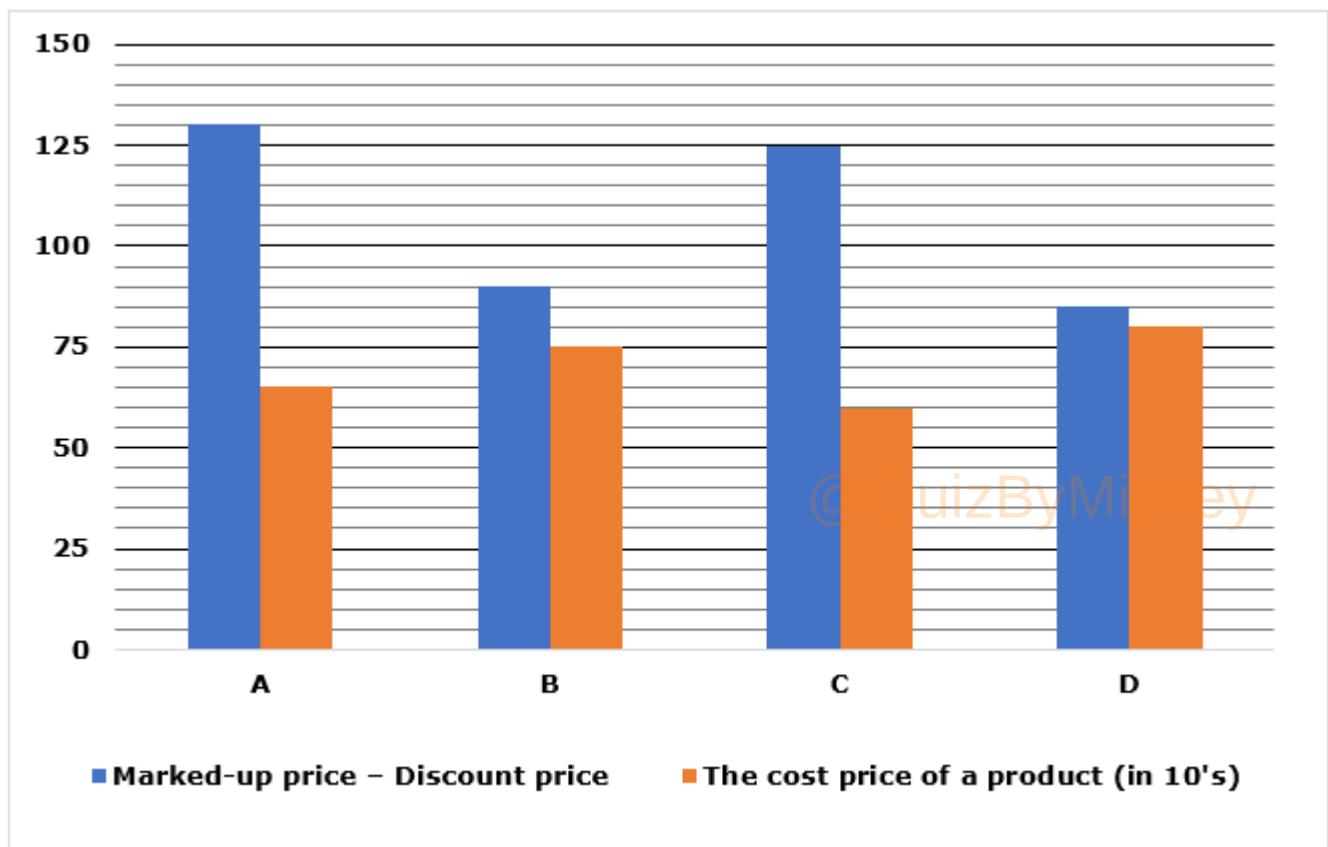
If the total number of male passengers in five trains together is 2196. Find the total number of female passengers in five trains together.

- a. 1236
- b. 1459
- c. 1318
- d. 1673
- e. 1579

6. Questions

Study the following information carefully and answer the following questions.

The bar graph below shows the difference between the marked-up price and the discount price (marked-up price – discount price) and the cost price (Rs.) of four different products namely A, B, C and D.



Notes:

- i). The ratio of the marked price to the selling price of product A is 4:3, respectively.
- ii). The marked-up percentage of product B is 40%, and the ratio of the marked-up price of products B and C is 10:9, respectively.
- iii). The discount percentage of product A and that of product D are the same.

If the discount price of product B is increased by Rs. 26. Then the new profit price of product B is what percentage of the profit price of product C.

- a. 51.2%
- b. 47.2%
- c. 57.2%
- d. 62.2%
- e. 41.2%

7. Questions

If the cost price of the product C is increased by 5% and the marked price and the profit percentage of the product C remains the same. Find the new discount percentage of product C.

- a. 25%
- b. 10%
- c. 17.5%
- d. 22.5%
- e. 12.5%

8. Questions

What is the ratio of the profit percentage of product A to the profit percentage of product B?

- a. 5: 3
- b. 3: 2
- c. 7: 3
- d. 5: 2
- e. 6: 1

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9. Questions

If the cost price of product E is 25% less than that of product D and it is marked up by 50% above the cost price. The successive discount of 20% and N% from the marked price of product E. The profit of product E, which is 14%. Which of the following statements is true?

I: N is a prime number

II: $(N + 4)$ is a perfect square

III: N is a multiple of 10.

- a. Both II and III
- b. Both I and III
- c. Only II
- d. Both I and II
- e. Only I

10. Questions

The difference between the profit and discount price of product A is Rs. P , and the difference between the profit and discount price of product D is Rs. S . Find the relationship between P and S .

- a. $P > S$
- b. $P < S$
- c. $(P + 60) = (S - 20)$
- d. Both (a) and (c)
- e. Both (b) and (c)

11. Questions

One number is wrong, and one number is missing in the given number series. Solve the number series and answer the questions given below.

I: 0.25, 2.5, 30, 240, 1440, 5760, P

Find the wrong number in the series.

- a. 1440
- b. 2.5
- c. 0.25
- d. 5760
- e. 30

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12. Questions

If the wrong number of the series is A , then find which of the statements is/are correct.

I: $P - 2880 = 3456A$

II: P is completely divisible by A

III: $P > 4750A$

- a. Both I and II
- b. All I, II and III
- c. None is true
- d. Both II and III
- e. Both I and III

13. Questions

Find the pattern of the given series and answer the question given below. The two given series follow different pattern.

Series I: 932, A, 972.5, 1004, 1049, 1112

Series II: (A/10), 67, 105, 57, 115, B

Find the value of $(A + B + 3)^{1/3}$.

- a. 8
- b. 12
- c. 11
- d. 10
- e. 5

14. Questions

Find the pattern of the series and answer the questions given below. The given three series follow different pattern.

Series I: a, d, c, 449, 613, 797, 1001

Series II: b, b+18, 236, 254, 272, 290

Series III: 256, 272, 208, 352, z

Which of the following statements is true?

- a. $(b + d - c) < z$
- b. $(b + d - c) = (a - 1)$
- c. $(z + a) > b$
- d. Both (b) and (c)
- e. Both (a) and (b)

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15. Questions

Which of the following statements is/are incorrect?

I: d is a prime number.

II: a and z are both odd numbers.

III: $(c - 4)$ is divided by 7, then the quotient is 21.5% of b .

- a. Only II
- b. Only I
- c. Only III
- d. Both I and III
- e. Both II and III

16. Questions

Solve the given two equations, I and II, and from the roots of these equations, find which of the new equations is formed. For making the equation, one root is taken from I and the other root is taken from II.

I: $x^2 - 12x + 32 = 0$

II: $y^2 - 32y + 256 = 0$

- a. $a^2 - 20a + 128 = 0$
- b. $b^2 - 24b + 64 = 0$
- c. $c^2 - 24c + 128 = 0$
- d. $d^2 - 20d + 36 = 0$
- e. $e^2 - 32e + 30 = 0$

17. Questions

Solve the quadratic equation given below and answer the question based on these equations.

I: $2x^2 - Px + 15 = 0$

II: $4y^2 - Ry + 15 = 0$

Notes:

- i). Roots of the equation I are A and B. The sum of roots is 5.5.
- ii). Roots of the equation II are C and D. The smallest root (C) of the equation II is $\frac{1}{2}$ of the smallest root (A) of the equation I.

Find the sum of the A and D.

- a. 4.5

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- b. 6.5
- c. 7
- d. 6
- e. 5.5

18. Questions

Which of the following statements is/are correct?

I: $A + 0.5 = D$

II: $C + 1.75 = D$

III: $D - C = B - 1.25$

- a. All I, II, and III
- b. Both I and III
- c. Both II and III
- d. None is true
- e. Both I and II

19. Questions

Find the value of M.

I: $Y^2 - 15Y + (7M^3) = 0$

Notes:

- i). The positive root of the equation is 4M.
- ii). M is a positive number.

- a. 3
- b. 1
- c. 2
- d. Cannot be determined
- e. 4

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20. Questions

I: $x^2 + Mx - 126 = 0$

II: $y^2 - My - 104 = 0$

Notes: The value of M and the positive root of equation II are prime numbers.

Which of the following statements is true?

P: The largest roots of both equations are prime numbers.

Q: The sum of the smallest root of both equations is -22.

R: The difference between the largest roots of the two equations is a composite number.

- a. Only P
- b. Only R
- c. Only Q
- d. None is true
- e. Both (b) and (c)

Explanations:

1. Questions

Train A,

The total number of passengers in a train = $3X * 50 = 150X$

The number of male passengers = 66% of $150X = 99X$

The number of female passengers = $150X - 99X = 51X$

Train B,

The total number of passengers in a train = $20 * 8X = 160X$

The number of male passengers = 54% of $160X = 86.4X$

The number of female passengers = $160X - 86.4X = 73.6X$

Let,

$$73.6X - 51X = 113$$

$$22.6X = 113$$

$$X = 5$$

So,

Train A,

The total number of passengers in a train = $150 * 5 = 750$

The number of male passengers = $99 * 5 = 495$

The number of female passengers = $750 - 495 = 255$

Train B,

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The total number of passengers in a train = $160 * 5 = 800$

The number of male passengers = 54% of $800 = 432$

The number of female passengers = $800 - 432 = 368$

Train D,

The total number of passengers in a train = $12 * 15X = 180X = 180 * 5 = 900$

The number of male passengers = 72% of $900 = 648$

The number of female passengers = $900 - 648 = 252$

Train E,

The total number of passengers in a train = $10 * 60 = 600$

The number of male passengers = 60% of $600 = 360$

The number of female passengers = $600 - 360 = 240$

Let table,

Trains	The number of coaches	The number of passengers in each coach	The total number of passengers	The number of male passengers	The number of female passengers
A	15	50	750	495	255
B	20	40	800	432	368
C	25	-	-	-	-
D	12	75	900	648	252
E	10	60	600	360	240

Answer: B

Let according to the question,

Seat capacity of new coach = $15X = 15 * 5 = 75$

The total number of passengers in train A now = $750 + 75 * 2 = 900$

The number of male passengers in train A now = 120% of $495 = 594$

The number of female passengers in train A now = $900 - 594 = 306$

Thus, required ratio = $594 : 306 = 33 : 17$

2. Questions

Train A,

The total number of passengers in a train = $3X * 50 = 150X$

The number of male passengers = 66% of $150X = 99X$

The number of female passengers = $150X - 99X = 51X$

Train B,

The total number of passengers in a train = $20 * 8X = 160X$

The number of male passengers = 54% of $160X = 86.4X$

The number of female passengers = $160X - 86.4X = 73.6X$

Let,

$$73.6X - 51X = 113$$

$$22.6X = 113$$

$$X = 5$$

So,

Train A,

The total number of passengers in a train = $150 * 5 = 750$

The number of male passengers = $99 * 5 = 495$

The number of female passengers = $750 - 495 = 255$

Train B,

The total number of passengers in a train = $160 * 5 = 800$

The number of male passengers = 54% of $800 = 432$

The number of female passengers = $800 - 432 = 368$

Train D,

The total number of passengers in a train = $12 * 15X = 180X = 180 * 5 = 900$

The number of male passengers = 72% of $900 = 648$

The number of female passengers = $900 - 648 = 252$

Train E,

The total number of passengers in a train = $10 * 60 = 600$

The number of male passengers = 60% of $600 = 360$

The number of female passengers = $600 - 360 = 240$

Let table,

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Trains	The number of coaches	The number of passengers in each coach	The total number of passengers	The number of male passengers	The number of female passengers
A	15	50	750	495	255
B	20	40	800	432	368
C	25	-	-	-	-
D	12	75	900	648	252
E	10	60	600	360	240

Answer: A

The number of passengers in each coach in train C = C

The total number of passengers in train C = $25 * C = 25C$

The number of male passengers in train C = 36% of $25C = 9C$

The number of female passengers in train C = $25C - 9C = 16C$

Let,

$$16C - 9C = 217$$

$$7C = 217$$

$$C = 31$$

The number of passengers in each coach in train C = 31

Thus, percentage = $(31/40) * 100 = 77.5\%$

3. Questions

Train A,

The total number of passengers in a train = $3X * 50 = 150X$

The number of male passengers = 66% of $150X = 99X$

The number of female passengers = $150X - 99X = 51X$

Train B,

The total number of passengers in a train = $20 * 8X = 160X$

The number of male passengers = 54% of $160X = 86.4X$

The number of female passengers = $160X - 86.4X = 73.6X$

Let,

$$73.6X - 51X = 113$$

$$22.6X = 113$$

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$$X = 5$$

So,

Train A,

$$\text{The total number of passengers in a train} = 150 * 5 = 750$$

$$\text{The number of male passengers} = 99 * 5 = 495$$

$$\text{The number of female passengers} = 750 - 495 = 255$$

Train B,

$$\text{The total number of passengers in a train} = 160 * 5 = 800$$

$$\text{The number of male passengers} = 54\% \text{ of } 800 = 432$$

$$\text{The number of female passengers} = 800 - 432 = 368$$

Train D,

$$\text{The total number of passengers in a train} = 12 * 15X = 180X = 180 * 5 = 900$$

$$\text{The number of male passengers} = 72\% \text{ of } 900 = 648$$

$$\text{The number of female passengers} = 900 - 648 = 252$$

Train E,

$$\text{The total number of passengers in a train} = 10 * 60 = 600$$

$$\text{The number of male passengers} = 60\% \text{ of } 600 = 360$$

$$\text{The number of female passengers} = 600 - 360 = 240$$

Let table,

Trains	The number of coaches	The number of passengers in each coach	The total number of passengers	The number of male passengers	The number of female passengers
A	15	50	750	495	255
B	20	40	800	432	368
C	25	-	-	-	-
D	12	75	900	648	252
E	10	60	600	360	240

Answer: C

$$\text{The number of AC coaches in train B} = (20/5) * 1 = 4$$

$$\text{The number of sleeper coaches in train B} = 20 - 4 = 16$$

$$\text{The total number of passengers in AC coaches in train B} = 4 * 40 = 160$$

The total number of passengers in sleeper coaches in train B = $16 * 40 = 640$

The number of male passengers in AC coaches in train B = $5X\%$ of $432 = 25\%$ of $432 = 108$

The number of male passengers in sleeper coaches in train B = $432 - 108 = 324$

The number of female passengers in sleeper coaches in train B = $640 - 324 = 316$

4. Questions

Train A,

The total number of passengers in a train = $3X * 50 = 150X$

The number of male passengers = 66% of $150X = 99X$

The number of female passengers = $150X - 99X = 51X$

Train B,

The total number of passengers in a train = $20 * 8X = 160X$

The number of male passengers = 54% of $160X = 86.4X$

The number of female passengers = $160X - 86.4X = 73.6X$

Let,

$$73.6X - 51X = 113$$

$$22.6X = 113$$

$$X = 5$$

So,

Train A,

The total number of passengers in a train = $150 * 5 = 750$

The number of male passengers = $99 * 5 = 495$

The number of female passengers = $750 - 495 = 255$

Train B,

The total number of passengers in a train = $160 * 5 = 800$

The number of male passengers = 54% of $800 = 432$

The number of female passengers = $800 - 432 = 368$

Train D,

The total number of passengers in a train = $12 * 15X = 180X = 180 * 5 = 900$

The number of male passengers = 72% of $900 = 648$

The number of female passengers = $900 - 648 = 252$

Train E,

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The total number of passengers in a train = $10 * 60 = 600$

The number of male passengers = 60% of 600 = 360

The number of female passengers = $600 - 360 = 240$

Let table,

Trains	The number of coaches	The number of passengers in each coach	The total number of passengers	The number of male passengers	The number of female passengers
A	15	50	750	495	255
B	20	40	800	432	368
C	25	-	-	-	-
D	12	75	900	648	252
E	10	60	600	360	240

Answer: D

The total number of passengers in train C = $(100 + 5X)\%$ of 600 = 125% of 600 = 750

The number of passengers in each coach in train C = $750/25 = 30$

Thus, difference = $60 - 30 = 30 = 6X$

I: $6X = 6 * 5 = 30$ (true)

II: $30 = 30$ (true)

III: $X + 25 = 5 + 25 = 30$ (true)

5. Questions

Train A,

The total number of passengers in a train = $3X * 50 = 150X$

The number of male passengers = 66% of $150X = 99X$

The number of female passengers = $150X - 99X = 51X$

Train B,

The total number of passengers in a train = $20 * 8X = 160X$

The number of male passengers = 54% of $160X = 86.4X$

The number of female passengers = $160X - 86.4X = 73.6X$

Let,

$$73.6X - 51X = 113$$

$$22.6X = 113$$

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$$X = 5$$

So,

Train A,

$$\text{The total number of passengers in a train} = 150 * 5 = 750$$

$$\text{The number of male passengers} = 99 * 5 = 495$$

$$\text{The number of female passengers} = 750 - 495 = 255$$

Train B,

$$\text{The total number of passengers in a train} = 160 * 5 = 800$$

$$\text{The number of male passengers} = 54\% \text{ of } 800 = 432$$

$$\text{The number of female passengers} = 800 - 432 = 368$$

Train D,

$$\text{The total number of passengers in a train} = 12 * 15X = 180X = 180 * 5 = 900$$

$$\text{The number of male passengers} = 72\% \text{ of } 900 = 648$$

$$\text{The number of female passengers} = 900 - 648 = 252$$

Train E,

$$\text{The total number of passengers in a train} = 10 * 60 = 600$$

$$\text{The number of male passengers} = 60\% \text{ of } 600 = 360$$

$$\text{The number of female passengers} = 600 - 360 = 240$$

Let table,

Trains	The number of coaches	The number of passengers in each coach	The total number of passengers	The number of male passengers	The number of female passengers
A	15	50	750	495	255
B	20	40	800	432	368
C	25	-	-	-	-
D	12	75	900	648	252
E	10	60	600	360	240

Answer: E

$$\text{The total number of male passengers in five trains together} = 2196$$

$$\text{The number of male passengers in train C} = 2196 - 495 - 432 - 648 - 360 = 261$$

$$\text{The number of female passengers in train C} = (261/36) * 64 = 464$$

The total number of female passengers in five trains together = $255 + 368 + 464 + 252 + 240 = 1579$

6. Questions

Product A,

The C.P = Rs. 650

The M.P = Rs. $4x$

The S.P = Rs. $3x$

The marked-up price = Rs. $(4x - 650)$

The discount price = $4x - 3x = \text{Rs. } x$

marked-up price – discount price = 130

$$4x - 650 - x = 130$$

$$3x = 780$$

$$x = 260$$

The M.P = $4 * 260 = \text{Rs. } 1040$

The S.P = $3 * 260 = \text{Rs. } 780$

The profit price = $780 - 650 = \text{Rs. } 130$

The discount price = $1040 - 780 = \text{Rs. } 260$

Product B,

The C.P = Rs. 750

The marked-up percentage of product B is 40%

The M.P = 140% of 750 = Rs. 1050

The marked-up price = $1050 - 750 = \text{Rs. } 300$

marked-up price – discount price = 90

$$300 - \text{discount price} = 90$$

The discount price = $300 - 90 = \text{Rs. } 210$

The S.P = $1050 - 210 = \text{Rs. } 840$

The profit price = $840 - 750 = \text{Rs. } 90$

Product C,

The C.P = Rs. 600

The ratio of the marked-up price of products B and C is 10:9, respectively

The marked-up price = $(300/10) * 9 = \text{Rs. } 270$

marked-up price – discount price = 125

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$$270 - \text{discount price} = 125$$

$$\text{The discount price} = 270 - 125 = \text{Rs. } 145$$

$$\text{The M.P} = 600 + 270 = \text{Rs. } 870$$

$$\text{The S.P} = 870 - 145 = \text{Rs. } 725$$

$$\text{The profit price} = 725 - 600 = \text{Rs. } 125$$

Product D,

$$\text{The C.P} = \text{Rs. } 800$$

The discount percentage of product A and that of product D are the same.

$$\text{The discount percentage of product A} = (260/1040) * 100 = 25\%$$

So, The discount percentage of product D is 25%

$$\text{The M.P} = \text{Rs. } D$$

$$\text{The marked-up price} = D - 800$$

$$\text{The discount price} = 25\% \text{ of } D = 0.25D$$

$$\text{marked-up price} - \text{discount price} = 85$$

$$D - 800 - 0.25D = 85$$

$$0.75D = 885$$

$$D = 1180$$

$$\text{The M.P} = \text{Rs. } 1180$$

$$\text{The S.P} = 75\% \text{ of } 1180 = \text{Rs. } 885$$

$$\text{The discount price} = 0.25 * 1180 = \text{Rs. } 295$$

$$\text{The profit price} = 885 - 800 = \text{Rs. } 85$$

Product	The C.P (Rs.)	The M.P (Rs.)	The S.P (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	650	1040	780	130	260
B	750	1050	840	90	210
C	600	870	725	125	145
D	800	1180	885	85	295

Answer: A

Let according to the question,

$$\text{The new S.P of product B} = 840 - 26 = \text{Rs. } 814$$

$$\text{The new profit price of product B} = 814 - 750 = \text{Rs. } 64$$

Thus, required percentage = $(64/125) * 100 = 51.2\%$

7. Questions

Product A,

The C.P = Rs. 650

The M.P = Rs. $4x$

The S.P = Rs. $3x$

The marked-up price = Rs. $(4x - 650)$

The discount price = $4x - 3x = \text{Rs. } x$

marked-up price – discount price = 130

$$4x - 650 - x = 130$$

$$3x = 780$$

$$x = 260$$

The M.P = $4 * 260 = \text{Rs. } 1040$

The S.P = $3 * 260 = \text{Rs. } 780$

The profit price = $780 - 650 = \text{Rs. } 130$

The discount price = $1040 - 780 = \text{Rs. } 260$

Product B,

The C.P = Rs. 750

The marked-up percentage of product B is 40%

The M.P = 140% of 750 = Rs. 1050

The marked-up price = $1050 - 750 = \text{Rs. } 300$

marked-up price – discount price = 90

$$300 - \text{discount price} = 90$$

The discount price = $300 - 90 = \text{Rs. } 210$

The S.P = $1050 - 210 = \text{Rs. } 840$

The profit price = $840 - 750 = \text{Rs. } 90$

Product C,

The C.P = Rs. 600

The ratio of the marked-up price of products B and C is 10:9, respectively

The marked-up price = $(300/10) * 9 = \text{Rs. } 270$

marked-up price – discount price = 125

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$$270 - \text{discount price} = 125$$

$$\text{The discount price} = 270 - 125 = \text{Rs. } 145$$

$$\text{The M.P} = 600 + 270 = \text{Rs. } 870$$

$$\text{The S.P} = 870 - 145 = \text{Rs. } 725$$

$$\text{The profit price} = 725 - 600 = \text{Rs. } 125$$

Product D,

$$\text{The C.P} = \text{Rs. } 800$$

The discount percentage of product A and that of product D are the same.

$$\text{The discount percentage of product A} = (260/1040) * 100 = 25\%$$

So, The discount percentage of product D is 25%

$$\text{The M.P} = \text{Rs. } D$$

$$\text{The marked-up price} = D - 800$$

$$\text{The discount price} = 25\% \text{ of } D = 0.25D$$

$$\text{marked-up price} - \text{discount price} = 85$$

$$D - 800 - 0.25D = 85$$

$$0.75D = 885$$

$$D = 1180$$

$$\text{The M.P} = \text{Rs. } 1180$$

$$\text{The S.P} = 75\% \text{ of } 1180 = \text{Rs. } 885$$

$$\text{The discount price} = 0.25 * 1180 = \text{Rs. } 295$$

$$\text{The profit price} = 885 - 800 = \text{Rs. } 85$$

Product	The C.P (Rs.)	The M.P (Rs.)	The S.P (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	650	1040	780	130	260
B	750	1050	840	90	210
C	600	870	725	125	145
D	800	1180	885	85	295

Answer: E

$$\text{The new cost price of the product C} = 105\% \text{ of } 600 = \text{Rs. } 630$$

$$\text{The profit percentage of product C} = [(725 - 600)/600] * 100 = (125/6)\%$$

$$\text{The new selling price of the product C} = [100 + (125/6)]\% \text{ of } 630 = (725/600) * 630 = \text{Rs. } 761.25$$

The new discount percentage of product C = $[(870 - 761.25)/870] * 100 = (108.75/870) * 100 = 12.5\%$

8. Questions

Product A,

The C.P = Rs. 650

The M.P = Rs. $4x$

The S.P = Rs. $3x$

The marked-up price = Rs. $(4x - 650)$

The discount price = $4x - 3x = \text{Rs. } x$

marked-up price – discount price = 130

$$4x - 650 - x = 130$$

$$3x = 780$$

$$x = 260$$

The M.P = $4 * 260 = \text{Rs. } 1040$

The S.P = $3 * 260 = \text{Rs. } 780$

The profit price = $780 - 650 = \text{Rs. } 130$

The discount price = $1040 - 780 = \text{Rs. } 260$

Product B,

The C.P = Rs. 750

The marked-up percentage of product B is 40%

The M.P = 140% of 750 = Rs. 1050

The marked-up price = $1050 - 750 = \text{Rs. } 300$

marked-up price – discount price = 90

$$300 - \text{discount price} = 90$$

The discount price = $300 - 90 = \text{Rs. } 210$

The S.P = $1050 - 210 = \text{Rs. } 840$

The profit price = $840 - 750 = \text{Rs. } 90$

Product C,

The C.P = Rs. 600

The ratio of the marked-up price of products B and C is 10:9, respectively

The marked-up price = $(300/10) * 9 = \text{Rs. } 270$

marked-up price – discount price = 125

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$$270 - \text{discount price} = 125$$

$$\text{The discount price} = 270 - 125 = \text{Rs. } 145$$

$$\text{The M.P} = 600 + 270 = \text{Rs. } 870$$

$$\text{The S.P} = 870 - 145 = \text{Rs. } 725$$

$$\text{The profit price} = 725 - 600 = \text{Rs. } 125$$

Product D,

$$\text{The C.P} = \text{Rs. } 800$$

The discount percentage of product A and that of product D are the same.

$$\text{The discount percentage of product A} = (260/1040) * 100 = 25\%$$

So, The discount percentage of product D is 25%

$$\text{The M.P} = \text{Rs. } D$$

$$\text{The marked-up price} = D - 800$$

$$\text{The discount price} = 25\% \text{ of } D = 0.25D$$

$$\text{marked-up price} - \text{discount price} = 85$$

$$D - 800 - 0.25D = 85$$

$$0.75D = 885$$

$$D = 1180$$

$$\text{The M.P} = \text{Rs. } 1180$$

$$\text{The S.P} = 75\% \text{ of } 1180 = \text{Rs. } 885$$

$$\text{The discount price} = 0.25 * 1180 = \text{Rs. } 295$$

$$\text{The profit price} = 885 - 800 = \text{Rs. } 85$$

Product	The C.P (Rs.)	The M.P (Rs.)	The S.P (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	650	1040	780	130	260
B	750	1050	840	90	210
C	600	870	725	125	145
D	800	1180	885	85	295

Answer: A

$$\text{The profit percentage of product A} = [(780 - 650)/650] * 100 = 20\%$$

$$\text{The profit percentage of product B} = [(840 - 750)/750] * 100 = 12\%$$

$$\text{Thus, required ratio} = 20: 12 = 5: 3$$

9. Questions

Product A,

The C.P = Rs. 650

The M.P = Rs. $4x$

The S.P = Rs. $3x$

The marked-up price = Rs. $(4x - 650)$

The discount price = $4x - 3x = \text{Rs. } x$

marked-up price – discount price = 130

$$4x - 650 - x = 130$$

$$3x = 780$$

$$x = 260$$

The M.P = $4 * 260 = \text{Rs. } 1040$

The S.P = $3 * 260 = \text{Rs. } 780$

The profit price = $780 - 650 = \text{Rs. } 130$

The discount price = $1040 - 780 = \text{Rs. } 260$

Product B,

The C.P = Rs. 750

The marked-up percentage of product B is 40%

The M.P = 140% of 750 = Rs. 1050

The marked-up price = $1050 - 750 = \text{Rs. } 300$

marked-up price – discount price = 90

$$300 - \text{discount price} = 90$$

The discount price = $300 - 90 = \text{Rs. } 210$

The S.P = $1050 - 210 = \text{Rs. } 840$

The profit price = $840 - 750 = \text{Rs. } 90$

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Product C,

The C.P = Rs. 600

The ratio of the marked-up price of products B and C is 10:9, respectively

The marked-up price = $(300/10) * 9 = \text{Rs. } 270$

marked-up price – discount price = 125

$$270 - \text{discount price} = 125$$

The discount price = $270 - 125 = \text{Rs. } 145$

The M.P = $600 + 270 = \text{Rs. } 870$

The S.P = $870 - 145 = \text{Rs. } 725$

The profit price = $725 - 600 = \text{Rs. } 125$

Product D,

The C.P = Rs. 800

The discount percentage of product A and that of product D are the same.

The discount percentage of product A = $(260/1040) * 100 = 25\%$

So, The discount percentage of product D is 25%

The M.P = Rs. D

The marked-up price = $D - 800$

The discount price = 25% of D = $0.25D$

marked-up price – discount price = 85

$D - 800 - 0.25D = 85$

$0.75D = 885$

$D = 1180$

The M.P = Rs. 1180

The S.P = 75% of 1180 = Rs. 885

The discount price = $0.25 * 1180 = \text{Rs. } 295$

The profit price = $885 - 800 = \text{Rs. } 85$

Product	The C.P (Rs.)	The M.P (Rs.)	The S.P (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	650	1040	780	130	260
B	750	1050	840	90	210
C	600	870	725	125	145
D	800	1180	885	85	295

Answer: D

The C.P of product E = 75% of 800 = Rs. 600

The M.P of product E = 150% of 600 = Rs. 900

The S.P of product E = 14% of 600 = Rs. 684

Let,

$$900 * 80/100 * (100 - N)/100 = 684$$

$$100 - N = 95$$

$$N = 5$$

Let,

I: N is a prime number

5 is a prime number

This is satisfied

II: (N + 4) is a perfect square

$$5 + 4 = 9$$

9 is a perfect square

This is satisfied

III: N is a multiple of 10.

5 is not a multiple of 10

This is not satisfied

10. Questions

Product A,

The C.P = Rs. 650

The M.P = Rs. 4x

The S.P = Rs. 3x

The marked-up price = Rs. (4x – 650)

The discount price = 4x – 3x = Rs. x

marked-up price – discount price = 130

$$4x - 650 - x = 130$$

$$3x = 780$$

$$x = 260$$

The M.P = 4 * 260 = Rs. 1040

The S.P = 3 * 260 = Rs. 780

The profit price = 780 – 650 = Rs. 130

The discount price = 1040 – 780 = Rs. 260

Product B,

The C.P = Rs. 750

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The marked-up percentage of product B is 40%

The M.P = 140% of 750 = Rs. 1050

The marked-up price = 1050 – 750 = Rs. 300

marked-up price – discount price = 90

300 – discount price = 90

The discount price = 300 – 90 = Rs. 210

The S.P = 1050 – 210 = Rs. 840

The profit price = 840 – 750 = Rs. 90

Product C,

The C.P = Rs. 600

The ratio of the marked-up price of products B and C is 10:9, respectively

The marked-up price = $(300/10) * 9$ = Rs. 270

marked-up price – discount price = 125

270 – discount price = 125

The discount price = 270 – 125 = Rs. 145

The M.P = 600 + 270 = Rs. 870

The S.P = 870 – 145 = Rs. 725

The profit price = 725 – 600 = Rs. 125

Product D,

The C.P = Rs. 800

The discount percentage of product A and that of product D are the same.

The discount percentage of product A = $(260/1040) * 100$ = 25%

So, The discount percentage of product D is 25%

The M.P = Rs. D

The marked-up price = D – 800

The discount price = 25% of D = 0.25D

marked-up price – discount price = 85

D – 800 – 0.25D = 85

0.75D = 885

D = 1180

The M.P = Rs. 1180

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The S.P = 75% of 1180 = Rs. 885

The discount price = $0.25 * 1180 = \text{Rs. } 295$

The profit price = $885 - 800 = \text{Rs. } 85$

Product	The C.P (Rs.)	The M.P (Rs.)	The S.P (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	650	1040	780	130	260
B	750	1050	840	90	210
C	600	870	725	125	145
D	800	1180	885	85	295

Answer: E

The difference between the profit and discount price of product A = $260 - 130 = \text{Rs. } 130$

$P = 130$

The difference between the profit and discount price of product D = $295 - 85 = \text{Rs. } 210$

$S = 210$

a). $P > S$

Let,

$130 > 210$

This is not satisfied.

b). $P < S$

Let,

$130 < 210$

This is satisfied.

c). $(P + 60) = (S - 20)$

Let,

$130 + 60 = 210 - 20$

$190 = 190$

This is satisfied.

11. Questions

Answer: B

$0.25 * 12 = 3$

$3 * 10 = 30$

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$$30 * 8 = 240$$

$$240 * 6 = 1440$$

$$1440 * 4 = 5760$$

$$5760 * 2 = 11520$$

The wrong number of the series is 2.5

Correct term of the series 3

$$P = 11520$$

12. Questions

Answer: A

$$A = 2.5$$

$$\text{I: } P - 2880 = 3456A$$

Let,

$$11520 - 2880 = 3456 * 2.5$$

$$8640 = 8640$$

This is satisfied.

II: P is completely divisible by A

Let,

$$11520/2.5 = 4608$$

This is satisfied.

III: P > 4750A

$$11520 > (4750 * 2.5)$$

$$11520 > 11875$$

This is not satisfied.

13. Questions

Answer: D

Series I:

932	950	972.5	1004	1049	1112
+18	+22.5	+31.5	+45	+63	
	+4.5	+9	+13.5	+18	

So, A = 950

$$A/10 = 950/10 = 95$$

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Series II:

95	67	105	57	115	47
-28	+38	-48	+58	-68	

$B = 47$

Let,

$(A + B + 3)^{1/3} = (950 + 47 + 3)^{1/3} = (1000)^{1/3} = 10$

14. Questions

Answer: E

Series I: a, d, c, 449, 613, 797, 1001

$613 - 449 = 164$

$797 - 613 = 184$

$1001 - 797 = 204$

So, the logic is difference of difference is 20

77	181	305	449	613	797	1001
+104	+124	+144	+164	+184	+204	

$a = 77$

$d = 181$

$c = 305$

Series II: b, b+18, 236, 254, 272, 290

$292 - 272 = 18$

$272 - 254 = 18$

$254 - 236 = 18$

$b + 18 - b = 18$

So, difference is 18

200	218	236	254	272	290
+18	+18	+18	+18	+18	

$b = 200$

Series III,

256	272	208	352	96
+16	-64	+144	-256	
4^2	8^2	12^2	16^2	

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$$z = 96$$

Let according to the question,

$$a). (b + d - c) < z$$

Let,

$$(200 + 181 - 305) < 96$$

$$76 < 96$$

This is satisfied.

$$b). (b + d - c) = (a - 1)$$

Let,

$$(200 + 181 - 305) = 77 - 1$$

$$76 = 76$$

This is satisfied.

$$c). (z + a) > b$$

Let,

$$(96 + 77) > 200$$

$$173 > 200$$

This is not satisfied.

15. Questions

Answer: A

I: d is a prime number.

Let,

181 is a prime number

This is satisfied.

II: a and z both are odd numbers.

Let,

$$z = 96 \text{ (not an odd number)}$$

This is not satisfied.

III: $(c - 4)$ is divided by 7, then the quotient is 21.5% of b.

Let,

$$(305 - 4)/7 = 301/7 \text{ (43 as quotient)}$$

$$21.5\% \text{ of } 200 = 43$$

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This is satisfied.

16. Questions

Answer: C

$$\text{I: } x^2 - 12x + 32 = 0$$

Let,

$$x^2 - 8x - 4x + 32 = 0$$

$$x(x - 8) - 4(x - 8) = 0$$

$$(x - 8)(x - 4) = 0$$

$$x = +8 \text{ and } +4$$

$$\text{II: } y^2 - 32y + 256 = 0$$

Let,

$$y^2 - 16y - 16y + 256 = 0$$

$$y(y - 16) - 16(y - 16) = 0$$

$$(y - 16)(y - 16) = 0$$

$$y = +16, +16$$

Let,

$$\text{c). } c^2 - 24c + 128 = 0$$

$$c^2 - 16c - 8c + 128 = 0$$

$$c(c - 16) - 8(c - 16) = 0$$

$$(c - 16)(c - 8) = 0$$

$$c = +16 \text{ and } +8$$

So, option (c) is satisfied.

17. Questions

Answer: E

$$\text{I: } 2x^2 - Px + 15 = 0$$

$$P = 5.5 * 2$$

$$P = 11$$

Let,

$$2x^2 - 11x + 15 = 0$$

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$$2x^2 - 6x - 5x + 15 = 0$$

$$2x(x - 3) - 5(x - 3) = 0$$

$$(2x - 5)(x - 3) = 0$$

$$x = +5/2 \text{ and } +3$$

The smallest root of equation II = $1/2 * 5/2 = +5/4$

$$\text{II: } 4y^2 - Ry + 15 = 0$$

$$4(5/4)^2 - R(5/4) + 15 = 0$$

$$6.25 - 5R/4 + 15 = 0$$

$$21.25 = 5R/4$$

$$5R = 85$$

$$R = 17$$

$$4y^2 - 17y + 15 = 0$$

Let,

$$4y^2 - 12y - 5y + 15 = 0$$

$$4y(y - 3) - 5(y - 3) = 0$$

$$(4y - 5)(y - 3) = 0$$

$$y = +3 \text{ and } +5/4$$

Then,

$$A = 5/2$$

$$B = 3$$

$$C = 5/4$$

$$D = 3$$

Let according to the question,

$$A + D = 5/2 + 3 = 5.5$$

18. Questions

Answer: A

$$\text{I: } A + 0.5 = D$$

Let,

$$5/2 + 0.5 = 3$$

$$3 = 3$$

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This is satisfied.

$$\text{II: } C + 1.75 = D$$

$$5/4 + 1.75 = 3$$

$$3 = 3$$

This is satisfied.

$$\text{III: } D - C = B - 1.25$$

$$3 - 5/4 = 3 - 1.25$$

$$1.75 = 1.75$$

This is satisfied.

19. Questions

Answer: C

$$\text{I: } Y^2 - 15Y + (7M^3) = 0$$

The product of the roots = $(7M^3)$

The positive root = $4M$

So, the negative root = $\{(7M^3)/4M\} = (7/4)M^2$

The sum of the roots = 15

Now,

$$4M + (7/4)M^2 = 15$$

$$16M + 7M^2 = 60$$

$$7M^2 + 16M - 60 = 0$$

$$7M^2 - 14M + 30M - 60 = 0$$

$$7M(M - 2) + 30(M - 2) = 0$$

$$(7M + 30)(M - 2) = 0$$

$$(M - 2)(7M + 30) = 0$$

$$M = 2 \text{ and } -30/7$$

20. Questions

Answer: E

$$\text{II: } y^2 - My - 104 = 0$$

Let 104 is divided by prime numbers (2 and 13) only

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Case i:

The positive root of equation II is 2.

$$\text{Another root} = 104/2 = 52$$

$$M = 52 - 2 = 50$$

50 is not a prime number.

Case ii:

The positive root of equation II is 13.

$$\text{Another root} = 104/13 = 8$$

$$M = 13 - 8 = 5$$

5 is a prime number.

So,

$$\text{II: } y^2 - 5y - 104 = 0$$

$$y^2 - 13y + 8y - 104 = 0$$

$$y(y - 13) + 8(y - 13) = 0$$

$$(y + 8)(y - 13) = 0$$

$$y = +13 \text{ and } -8$$

$$\text{I: } x^2 + Mx - 126 = 0$$

$$x^2 + 5x - 126 = 0$$

$$x^2 + 14x - 9x - 126 = 0$$

$$x(x + 14) - 9(x + 14) = 0$$

$$(x - 9)(x + 14) = 0$$

$$x = +9 \text{ and } -14$$

P: The largest roots of both equations are prime numbers.

Let,

9 is not a prime number.

This is not satisfied.

Q: The sum of the smallest root of both equations is -22.

Let,

$$\text{Sum} = -14 - 8 = -22$$

This is satisfied.

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R: The difference between the largest roots of the two equations is a composite number.

Let,

Thus, difference = $13 - 9 = 4$ (composite number)

This is satisfied.

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